# Statistics for Chemical Engineers:
## From Data to Models to Decisions

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## Chapter 1: Introduction

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# Motivation

As engineers, we often use *fundamental laws* to make *decisions*:

- Discovery of fundamental laws has been the result of extensive collection and analysis of observations (data)
- Fundamental law is often expressed in the form of a mechanistic model
- Mechanistic model provides a concise summary of observations (knowledge) that allows us to predict and generalize

**Can you think of fundamental laws used in chemical engineering?**

# Motivation

Fundamental laws are powerful but only provide limited descriptions of phenomena:

- Laws are applicable under specific settings (e.g., continuum vs. atomistic)
- Discovering laws and new mechanistic models might be challenging or cost-prohibitive (e.g., climate)

To account for this, we also often build predictive models based purely on observations (data); such models are known as *empirical models* and also embed knowledge.

Engineering decisions rely on a combination of mechanistic and empirical knowledge.

**Can you think of empirical models used in chemical engineering?**

# Motivation

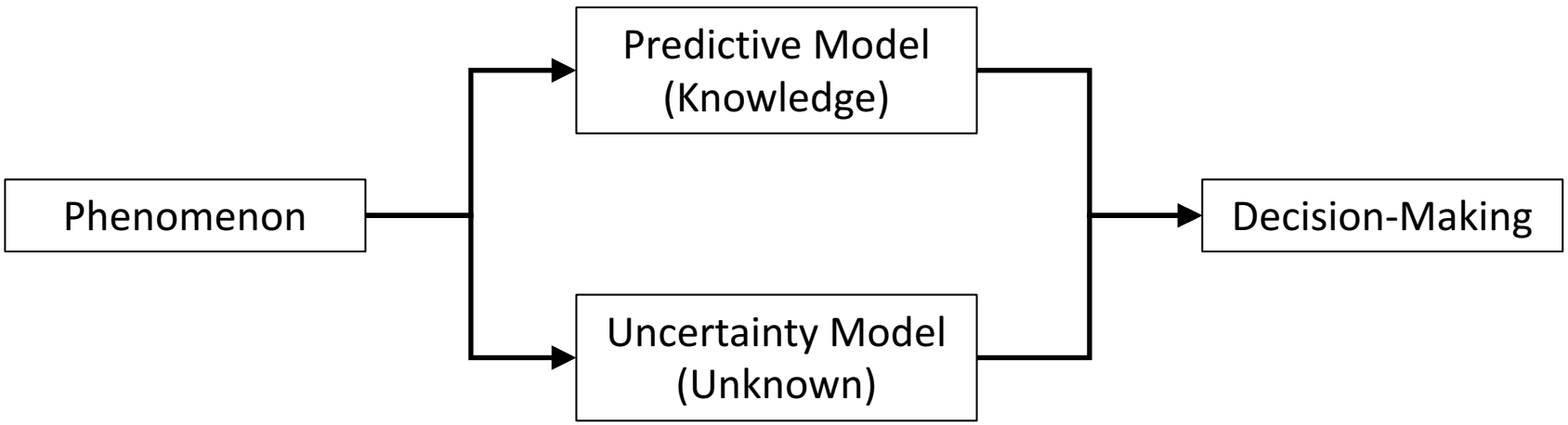
However:

- Model predictions will *always* face a certain degree of *uncertainty* (due to limited knowledge/understanding/data or due to *random* phenomena affecting systems).
- Despite these limitations, we still need to be able to *make decisions*. In fact, we (as humans) make predictions and decisions in our daily lives.
- The human brain naturally gathers knowledge (learns) from observations by building empirical models and has the ability to blend such empirical knowledge with mechanistic knowledge.
- The human brain has a natural ability to hedge against uncertainty and to adapt decisions based on new knowledge.

**Examples of random phenomena:** time to failure of a material, "pop" time of a corn kernel, molecular fluctuations.

**Examples of brain learning:** riding a bicycle, cooking.

# Motivation




Decision-making relies on the ability to characterize what is known (predictable) and not known (not predictable).

# Motivation

*Statistics* is the branch of mathematics that offers tools to:

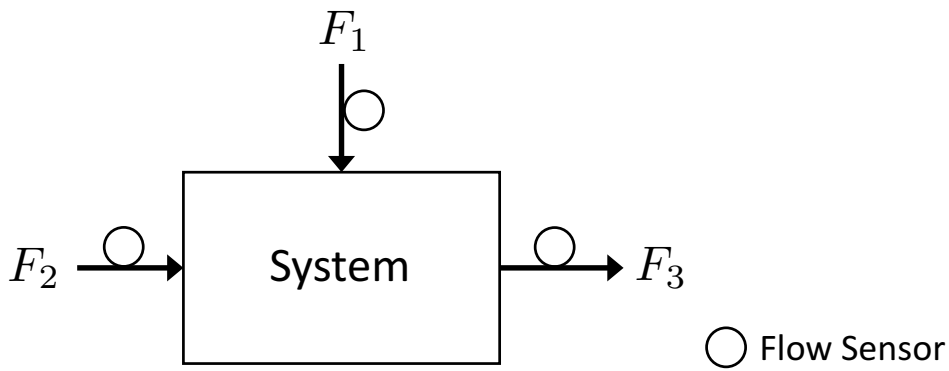
- Collect, analyze, and extract knowledge (models) from data
- Characterize and model the unknown (uncertainty)
- Systematically make decisions in the face of uncertainty

From an engineering perspective, *statistics* aids the discovery of fundamental laws and the development of mechanistic and empirical models as well as the characterization of random phenomena.

From a scientific perspective, *statistics* provides a framework for thinking about the world that can help us understand how humans extract knowledge from data and use this to make decisions in the face of uncertainty.

# Example: Flow Mixer `ch1_mixer_example` [matlab] [live] [python]

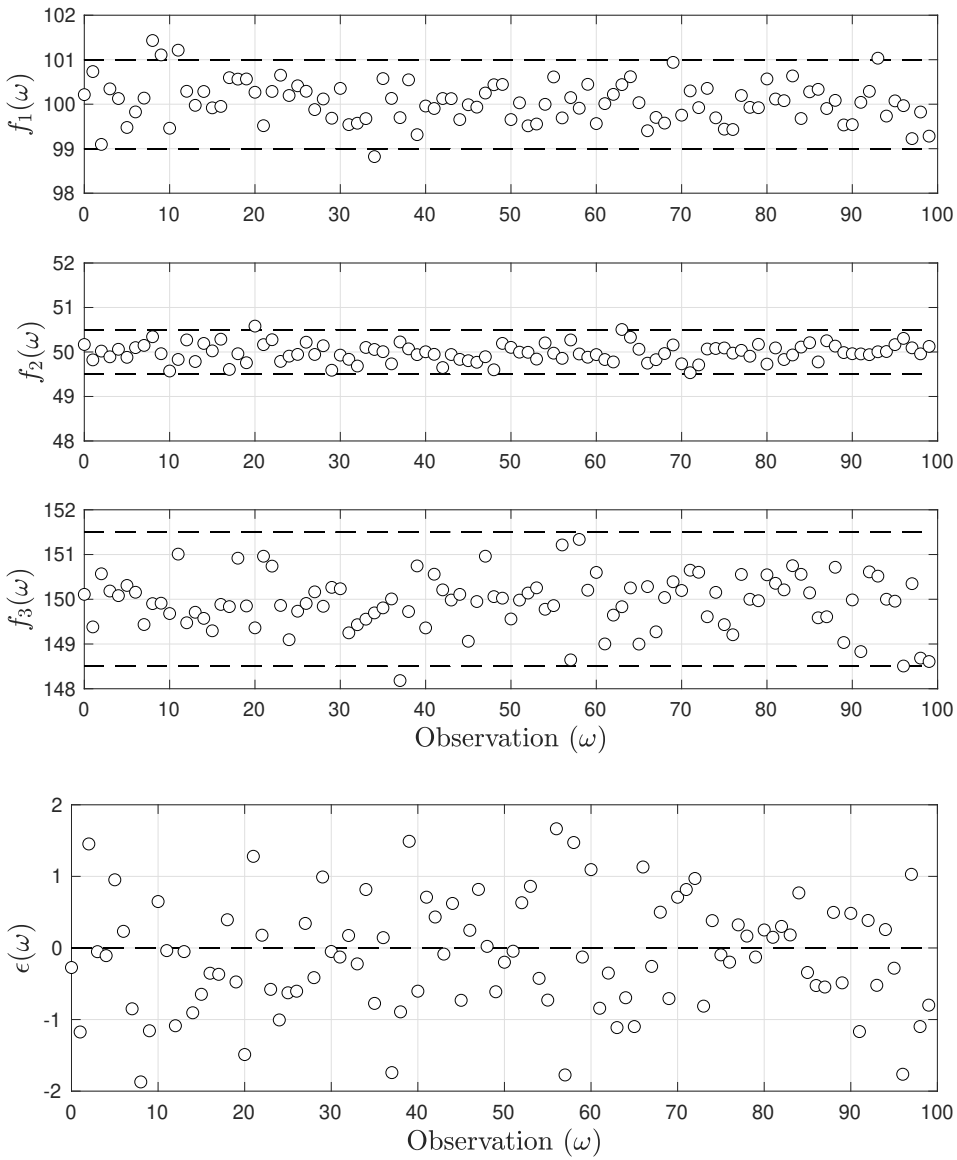
- Consider a mixing system with input flows $F_1$, $F_2$ and output flow $F_3$
- Sensors measure flows at time $\omega$: $f_1(\omega) = 101.5$, $f_2(\omega) = 50.5$, $f_3(\omega) = 151$ gpm
- Conservation laws tell us that $f_3(\omega) = f_1(\omega) + f_2(\omega)$ (but this is not true). Why?




- What is known and unknown about this system?
- What is the source of uncertainty?

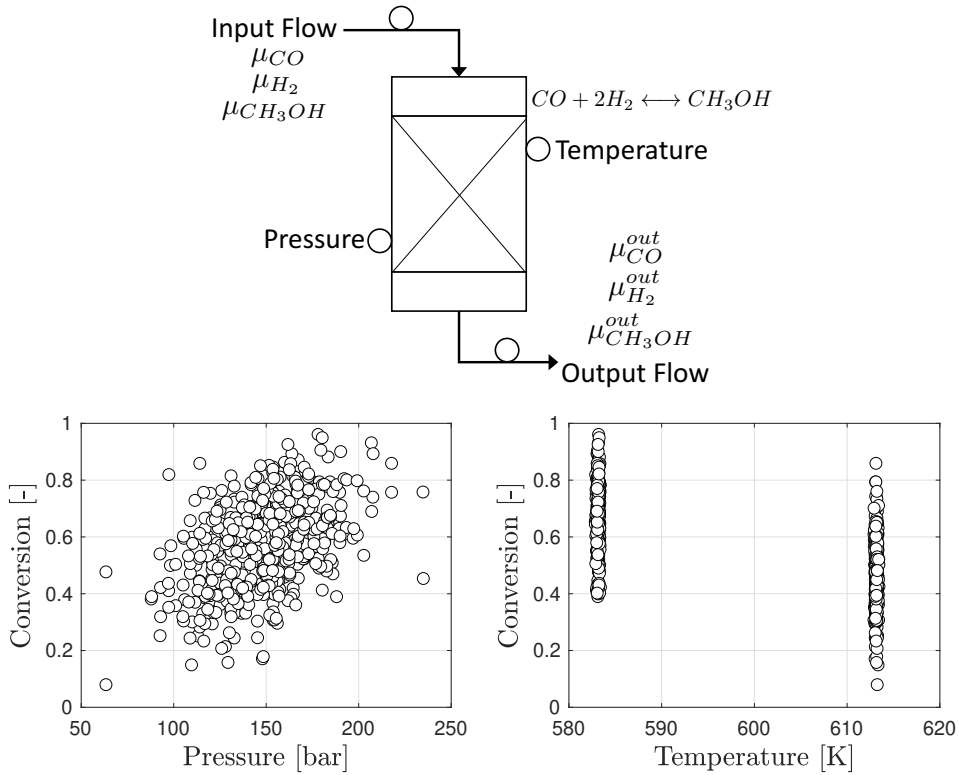
# Example: Flow Mixer `ch1_mixer_example` [matlab] [live] [python]

Collect 100 observations and monitor mismatch $\epsilon(\omega) = f_3(\omega) - f_1(\omega) - f_2(\omega)$



# Example: Gibbs Reactor

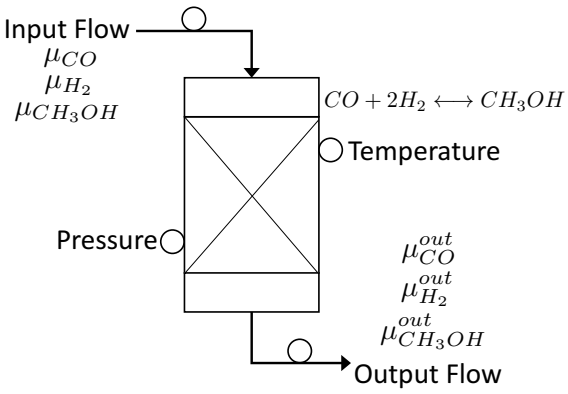
- Consider a reactor in which the reaction $CO + 2H_2 \leftrightarrow CH_3OH$ takes place




- Data seem to indicate that the reaction is favored (achieves higher conversion $\xi$) at high pressure ($P$) and low temperature ($T$) but this is blurred by variability.

- What is known and unknown about this system?
- Note that trends are blurred by uncertainty/noise.

# Example: Gibbs Reactor




$$\mu_k^{out} = \mu_k + \gamma_k \cdot \xi \cdot \mu_{CO},\ k \in K$$

$$\xi = 1 - \frac{\mu_{CO}^{out}}{\mu_{CO}}$$

$$K_{eq} = \frac{(\mu_{CH_3OH} + \xi \cdot \mu_{CO})}{(\mu_{CO} - \xi\mu_{CO})(\mu_{H_2} - 2 \cdot \xi \cdot \mu_{CO})^2} \left(\frac{\mu_{out}}{P}\right)^2$$

$$\frac{\partial \log K_{eq}}{\partial T} = \frac{\Delta H_0}{R \cdot T^2}$$

- We know fundamental conservation and thermodynamic laws hold and the mechanistic model allows us to predict $\xi$ from $P$ and $T$.
- This knowledge, however, is limited (e.g., makes assumptions, cannot explain random behavior).
- How to predict $\xi$ as a function of $P$ and $T$ without any mechanistic knowledge?
- What are trade-offs between mechanistic and empirical knowledge?

# Motivation

**Don't forget:**

Inherent limitation of engineering practice: no matter how sophisticated our predictive models and sensing devices are, we will *always* have a certain degree of uncertainty.

Characterizing both known and unknown aspects of a system is key.

# Random Variables

- In statistics, we use random variables (RVs) to *model* unknown (random) behavior.
- An RV does not have a known value and exhibits variability.
- The statistical view of the world is a significant departure from the traditional deterministic viewpoint (commonly used in engineering).
- Under a deterministic view of the world, we *assume* that variables have known and unique values (ignore uncertainty and variability).

- Deterministic Thinking: Temperature is $20^o$C (no uncertainty)
- Statistical Thinking: Temperature is $20 \pm 0.5^o$C (account for uncertainty)

# Random Variables

An RV model (denoted as $X$) has the following elements:

- Set of possible realizations $\omega \in \Omega$ with associated values $x_\omega \in \mathcal{D}_X$.
- Domain $\mathcal{D}_X$ under which realizations $x_\omega$ of $X$ "live".
- Probability measure $\mathbb{P} : \Omega \to [0, 1]$ that assigns probability to events.
- Cumulative density function (cdf) $F_X : \mathcal{D}_X \to [0, 1]$ that tells us $\mathbb{P}(X \leq t)$.
- Associated with cdf, there is a probability density function (pdf) $f_X : \mathcal{D}_X \to [0, \infty)$.

# Random Variables

Some observations:

- Think of $\omega$ as being a pinball that carries data $x_\omega$
- When you drop a pinball, this will "fall" in different locations of the domain $\mathcal{D}_X$
- pdf/cdf tell us how "densely" pinballs accumulate in certain locations of the domain.
- Where in the domain the pinball falls is an event and probabilities tell us how likely specific events are.
- An event is a subdomain $\mathcal{A} \subseteq \mathcal{D}_X$ (location or set of locations in domain).
- Probabilities are related to how densely pinballs accumulate in parts of domain.

# Random Variables

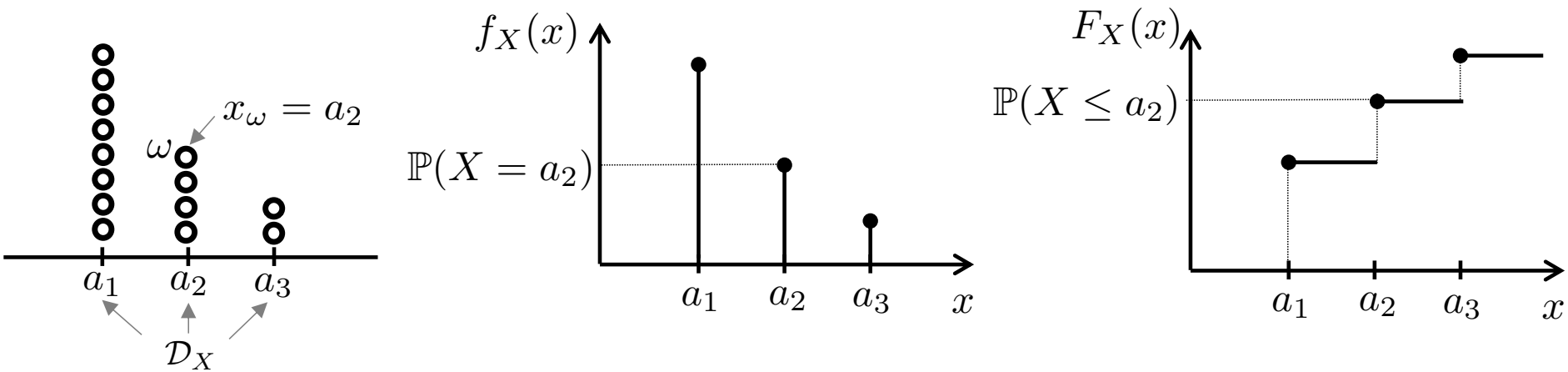



Figure: Illustration of the elements of a random variable.

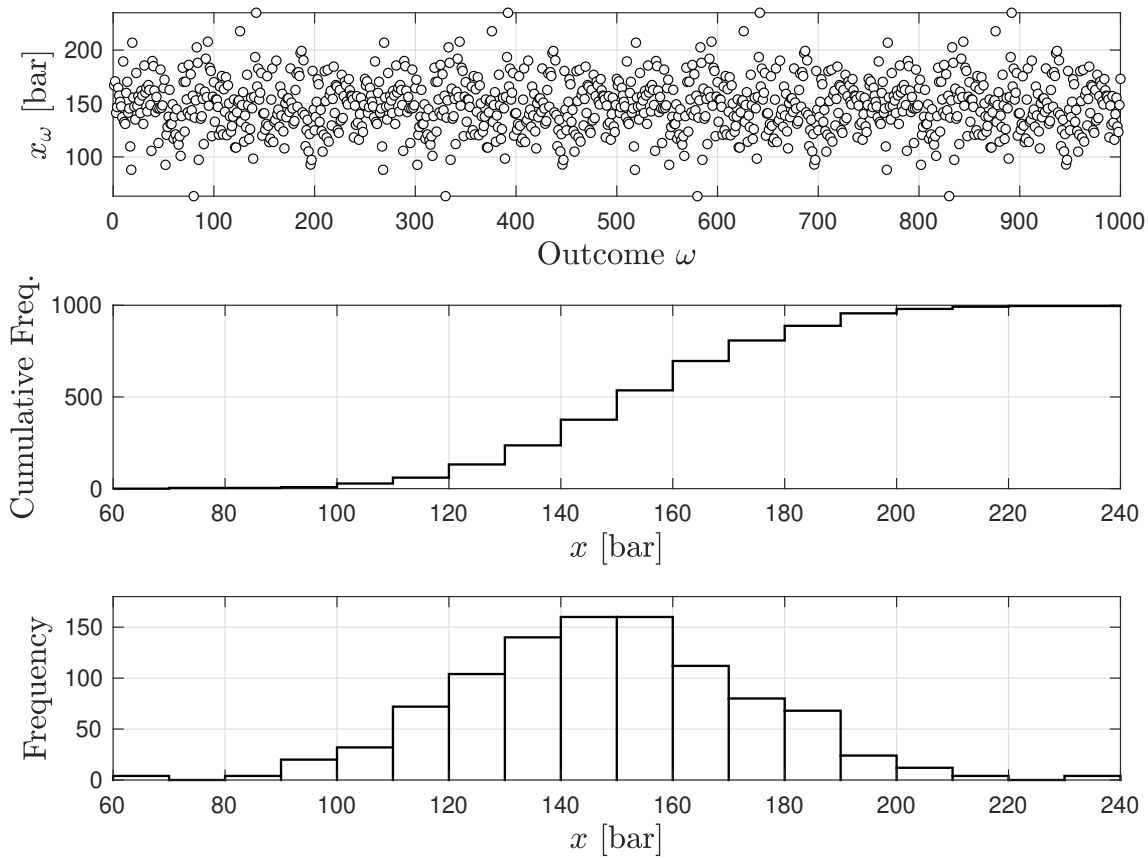
# Random Variables

**Don't Forget:**

- A random variable is a *model* (a characterization) of a random phenomenon.
- To fully define an RV model, you need to know $\mathcal{D}_X$, $f_X$, and $F_X$.

# Example: Gibbs Reactor `ch1_gibbs_example` [matlab] [live] [python]

- Assume *pressure* varies due to malfunction of flow control and model this as RV $X$
- Collect 1000 realizations $x_\omega$ (data); visualize as cumulative frequency and frequency.
- Cumulative frequency approximates cdf, frequency approximates pdf.



# Types of Random Variables

RVs are categorized as multivariate vs. univariate:

- A *multivariate* RV $X = (X_1, X_2, ..., X_n)$ has realizations that generate vector values $x_\omega = (x_{\omega,1}, x_{\omega,2}, ..., x_{\omega,n}) \in \mathbb{R}^n$.
- A *univariate* RV $X$ is a multivariate RV with $n = 1$ whose realizations are scalar values $x_\omega \in \mathbb{R}$.

- For a univariate RV, a pinball $\omega$ (observation) carries a single number $x_\omega$ (e.g., temperature).
- For a multivariate RV, a pinball $\omega$ carries a set of numbers $x_\omega = (x_{\omega,1}, x_{\omega,2}, ..., x_{\omega,n})$ (e.g., temperature, pressure, conversion)

For now, we will focus our discussion on univariate RVs.

# Types of Random Variables

RVs are categorized as continuous vs. discrete:

- A *continuous* RV $X$ is one in which the domain $\mathcal{D}_X$ is continuous; e.g., $X$ has realizations satisfying $0 \leq x_\omega \leq 1$.
- A *discrete* RV $X$ is one in which the domain $\mathcal{D}_X$ is discrete; e.g., $X$ has realizations satisfying $x_\omega \in \{0, 1\}$.

Many RV models are available that apply to different categories.

Type of RV used to characterize uncertainty depends on nature of random phenomenon.

# Probability Density of Discrete and Continuous RVs

Discrete $X$ has discrete domain $\mathcal{D}$ and cdf/pdf are discontinuous functions.

The pdf and cdf have the following properties:

1. $f(x) \geq 0, \quad x \in \mathcal{D}$
2. $f(x) = \mathbb{P}(X = x), \quad x \in \mathcal{D}$
3. $\sum_{x \in \mathcal{D}} f(x) = \sum_{x \in \mathcal{D}} \mathbb{P}(X = x) = 1$
4. $\mathbb{P}(X \in \mathcal{A}) = \sum_{x \in \mathcal{A}} f(x), \quad \mathcal{A} \subseteq \mathcal{D}.$
5. $F(t) = \mathbb{P}(X \leq t) = \sum_{x \in \mathcal{D} \mid x \leq t} f(x) \quad t \in \mathbb{R}.$

# Probability Density of Discrete and Continuous RVs

A discrete RV is easy to handle computationally (simple summations and counting):

- Since $\mathcal{D}$ is discrete, we have that:

$$\begin{aligned}\mathbb{P}(X \leq t) &= F(t) \\ &= \sum_{x \in \mathcal{D} | x \leq t} f(x) \\ &= \sum_{x \in \mathcal{D}} \mathbf{1}[x \leq t] f(x).\end{aligned}$$

  Here, we use the indicator function:

$$\mathbf{1}[x \leq t] = \begin{cases} 1 & \text{if} \quad x \leq t \\ 0 & \text{if} \quad x > t \end{cases}.$$

# Probability Density of Discrete and Continuous RVs

- Can compute pdf and cdf by *counting* how many times realizations $x_\omega$ of $X$ take a certain value (or are below a certain value):

$$f(x) = \mathbb{P}(X = x) = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \mathbf{1}[x_\omega = x]$$

$$F(t) = \mathbb{P}(X \leq t) = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \mathbf{1}[x_\omega \leq t].$$

- We can thus determine pdf and cdf of discrete RV $X$ directly from data $x_\omega$.

# Example Discrete RV

- Consider a discrete RV with domain $\mathcal{D} = \{-1, 0, 1, 2\}$ and pdf:

$$f(x) = \begin{cases} 0 & \text{for } x = -1 \\ 0.4 & \text{for } x = 0 \\ 0.6 & \text{for } x = 1 \\ 0 & \text{for } x = 2 \end{cases}$$

- Pdf satisfies $\sum_{x \in \mathcal{D}} f(x) = 1$ and $f(x) \geq 0$ for all $x \in \mathcal{D}$.
- RV has 10 possible realizations $\Omega = \{1, 2, 3, ..., 10\}$ with associated values $x_\omega$ given by $\{0, 1, 0, 0, 1, 1, 1, 1, 1, 0\}$.
- Pdf can also be computed using the observations, for example:

$$f(0) = \mathbb{P}(X = 0) = \frac{1}{10} \sum_{\omega \in \Omega} \mathbf{1}[x_\omega = 0] = 0.4$$

- Similarly, use observations to compute the cdf, for example:

$$F(0) = \mathbb{P}(X \leq 0) = \sum_{x | x \leq 0} f(x) = f(-1) + f(0) = 0.4$$

# Example Discrete RV

Pdf and cdf of the RV are:

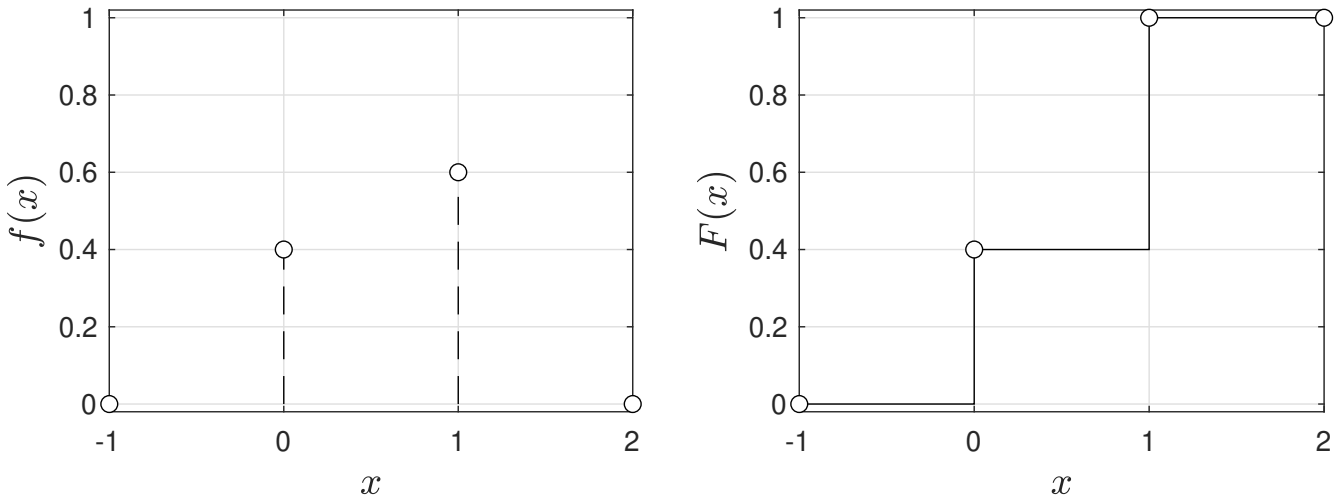



Figure: Pdf and cdf for simple discrete random variable.

Note discontinuous nature of these functions.

# Probability Density of Discrete and Continuous RVs

Continuous $X$ has continuous domain $\mathcal{D}$ and pdf/cdf are continuous functions.

The pdf and cdf have the following properties:

1. $f(x) \geq 0, \quad x \in \mathcal{D}.$
2. $f(x) = \frac{dF(x)}{dx}, \quad x \in \mathcal{D}.$
3. $\int_{x \in \mathcal{D}} f(x)dx = 1.$
4. $\mathbb{P}(X \in \mathcal{A}) = \int_{x \in \mathcal{A}} f(x)dx, \quad \mathcal{A} \subseteq \mathcal{D}.$
5. $F(t) = \int_{x \in \mathcal{D} | x \leq t} f(x)dx.$

# Probability Density of Discrete and Continuous RVs

- The 2nd property tells us that pdf $f(x)$ for a continuous RV $X$ is not $\mathbb{P}(X = x)$ (as in the discrete case).
- Instead, the pdf is the derivative of the cdf and thus:

$$\begin{aligned} f(x)\Delta x &\approx \mathbb{P}(X \leq x + \Delta x) - \mathbb{P}(X \leq x) \\ &= \mathbb{P}(x \leq X \leq x + \Delta x). \end{aligned}$$

  We thus have that the pdf tells us the probability that $X$ is in a *neighborhood* of $x$.

- Note that setting $\Delta x \to 0$ implies that $\mathbb{P}(X = x) = 0$.
- The 3rd, 4th, and 5th properties are analogous to the discrete case, but summations are replaced with integrals.

# Probability Density of Discrete and Continuous RVs

A continuous RV is more difficult to handle computationally since it involves integrals instead of summations (this prevents the use of simple counting operations).

- However, computations are analogous to those of the discrete case. For instance:

$$\begin{aligned}\mathbb{P}(X \leq t) &= \int_{x \in \mathcal{D} | x \leq t} f(x) dx \\ &= \int_{x \in \mathcal{D}} \mathbf{1}[x \leq t] f(x) dx.\end{aligned}$$

- Continuous RVs are often approximated using discretization. For instance:

$$f(x) \approx \frac{F(x + \Delta x) - F(x)}{\Delta x}.$$

- We will see that we can approximate continuous pdfs/cdfs from data $x_\omega$.

# Example Continuous RV

- Consider continuous RV with domain $\mathcal{D} = (-\infty, \infty)$ and pdf of the form:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

- It is easy to see that the pdf satisfies $f(x) \geq 0$ for any $x \in \mathcal{D}$ and one can show that $\int_{x \in \mathcal{D}} f(x)dx = 1$ (this is not that easy to show).

- The cdf associated with this pdf is given by:

$$\begin{aligned} F(t) &= \int_{x \leq t} f(x)dx = \int_{-\infty}^{t} f(x)dx \\ &= \frac{1}{2}\left(1 + \operatorname{erf}\left(\frac{t}{\sqrt{2}}\right)\right), \end{aligned}$$

- One can show $f(x) = dF(x)/dx$ and that cdf satisfies $0 \leq F(t) \leq 1$ for any $t \in \mathcal{D}$.

# Example Continuous RV

Pdf and cdf of the RV are:

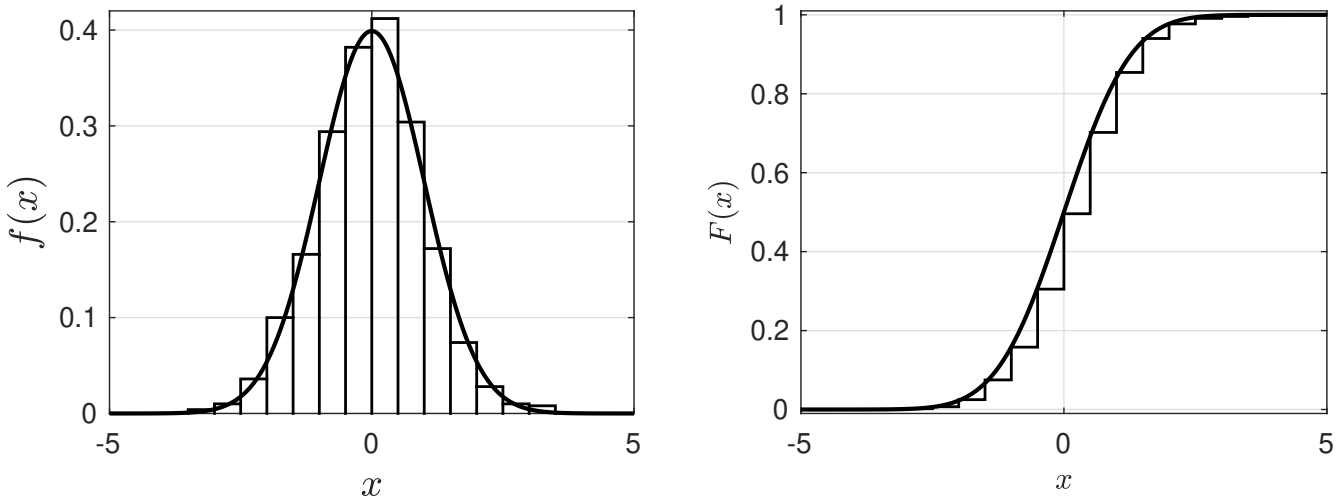



Figure: Pdf and cdf of continuous RV and corresponding discrete approximations.

Continuous pdf and cdf are approximated using discretized versions with $\Delta x = 0.5$.

# From Data to Models

- In practical situations, we have access to observations (data) of our system.
- Our goal is to use such data to gain knowledge (understanding) of our system.
- Our knowledge will be extracted from data in the form of *models* of two types:
  - *Structural models*: provide a characterization of structural dependencies between variables in our system (mechanistic or empirical)
  - *Random variable (uncertainty) models*: provide a characterization of behavior that cannot be explained by structural models

**Important:** Both models are necessary to make proper predictions and decisions.

# From Data to Models

- We will begin our discussion with random variable (RV) models.
- Assume that we have available observations (data) $x_\omega$, $\omega \in \mathcal{S}$.
- What type of an RV model are the observations following?
- How to obtain a cdf $F(x)$ and pdf $f(x)$ for an RV model from available data?

# From Data to Models

- Goal is to use available data $x_\omega$ to postulate a *theoretical* RV model $X$.
- RV $X$ is a *model* of a random phenomenon that generates the observed data.
- Many models are available to capture diverse phenomena seen in real life.
- A widely used model is that of a Gaussian RV.
- Model assumes that $X$ is continuous, has domain $\mathcal{D} = (-\infty, \infty)$, and has pdf:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathcal{D},$$

  where $\mu \in \mathbb{R}$ and $\sigma \in \mathbb{R}_+$ are parameters of the model.
- Parameters define behavior of the RV and can be "tuned" to match our data.
- The cdf associated with the Gaussian model is:

$$F(x) = \frac{1}{2}\left(1 + \operatorname{erf}\left(\frac{(x-\mu)/\sigma}{\sqrt{2}}\right)\right), \quad x \in \mathcal{D},$$

- **What other theoretical models do you know?**

# From Data to Models

- Our goal is to investigate whether the available observations $x_\omega$, $\omega \in \mathcal{S}$ follow the pdf and cdf of a theoretical model that we postulate.
- To verify this hypothesis, we use the data to construct *empirical* approximations for the pdf $f(x)$ and cdf $F(x)$ of the theoretical model and verify if these match.
- Empirical approximations (a.k.a. data-driven or sample-based approximations) are denoted as $\hat{f}(x)$ and $\hat{F}(x)$ and these are used to *estimate* $f(x)$ and $F(x)$.

# From Data to Models

The approach to construct $\hat{f}(x)$ and $\hat{F}(x)$ from data (for a continuous RV) can be summarized as follows:

1. Construct *empirical* domain $\hat{\mathcal{D}}$; this is the domain covered by observations $x_\omega$, $\omega \in \mathcal{S}$. This gives us an approximation of the domain $\mathcal{D}$ of the RV model. Discretize the domain $\hat{\mathcal{D}}$ into bins of size $\Delta x$.

2. Construct an empirical cdf:

$$\hat{F}(x) = \frac{1}{S} \sum_{\omega \in \mathcal{S}} \mathbf{1}[x_\omega \leq x], \quad x \in \hat{\mathcal{D}}$$

   This is the fraction of observations $x_\omega$ that take a value below $x$ (number normalized by $S$).

   The number of observations is known as the *cumulative frequency* and is given by:

$$\sum_{\omega \in \mathcal{S}} \mathbf{1}[x_\omega \leq x]$$

3. Construct the empirical pdf:

$$\hat{f}(x) = \frac{\frac{1}{S}\sum_{\omega \in \mathcal{S}} \mathbf{1}[x \leq x_\omega \leq x + \Delta x]}{\Delta x}$$

where $\Delta x$ is the size of the bin interval.

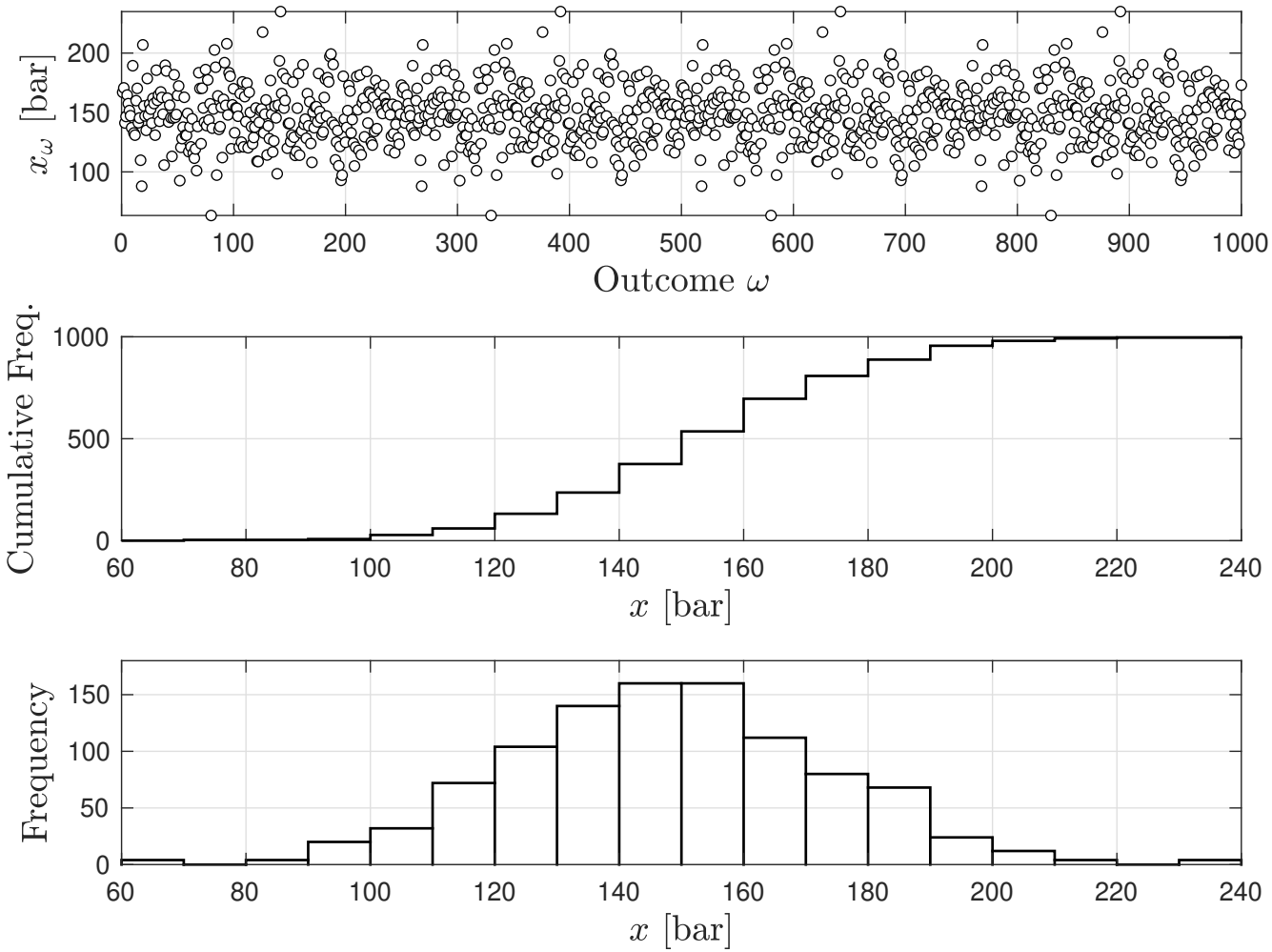
The number of observations in $[x, x + \Delta x]$ is known as the *frequency* and is given by:

$$\sum_{\omega \in \mathcal{S}} \mathbf{1}[x \leq x_\omega \leq x + \Delta x]$$

The procedure to obtain the empirical pdf/cdf of a discrete RV is easier, as the domain does not need to be discretized.

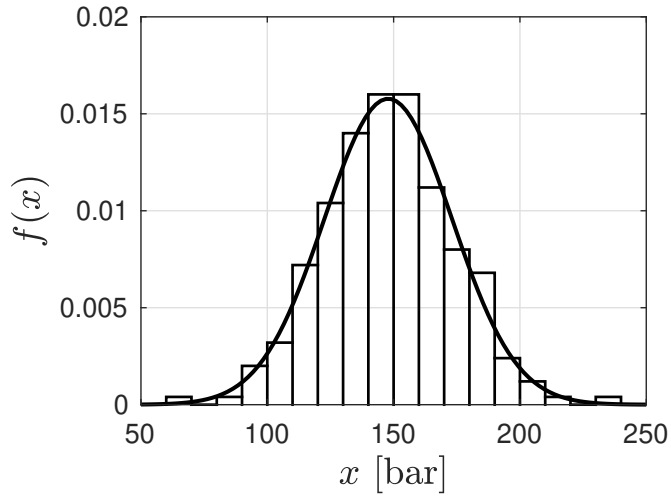
# Example: Gibbs Reactor `ch1_gibbs_example` [matlab] [live] [python]

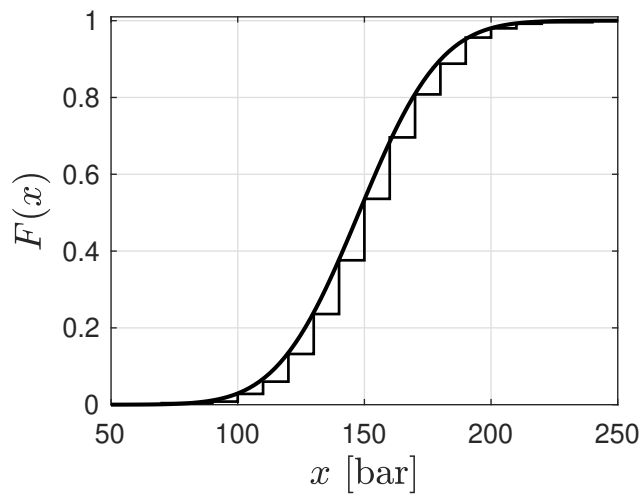
Here are $S$ =1,000 observations $x_\omega$ along with the frequency and cumulative frequency.



# Example: Gibbs Reactor `ch1_gibbs_example` [matlab] [live] [python]

- Obtain empirical cdf by normalizing cumulative frequency with $S = 1000$
- Obtain empirical pdf by normalizing frequency with $S = 1000$ and $\Delta x = 10$




- Empirical pdf and cdf match the pdf and cdf of a Gaussian RV model.
- We conclude that our random phenomenon (pressure) behaves as a Gaussian RV.

# Summarizing Statistics (Basic)

- Pdf and cdf are *functions* that fully characterize an RV $X$. However, in practice, we might be interested in using values (and not functions) to describe $X$.
- This is done by using *summarizing statistics* (a.k.a. descriptive statistics). Popular summarizing statistics are the expected value and variance:

For a discrete RV we have:

- *Expected Value (measure of magnitude):* $\mathbb{E}_X = \sum_{x \in \mathcal{D}_X} x f(x)$
- *Variance and Standard Deviation (measure of variability/uncertainty):*

$$\mathbb{V}_X = \sum_{x \in \mathcal{D}_X} f(x)(x - \mathbb{E}_X)^2, \qquad \mathbb{SD}_X = \sqrt{\mathbb{V}_X}$$

For a continuous RV we have:

- *Expected Value (measure of magnitude):* $\mathbb{E}_X = \int_{x \in \mathcal{D}_X} x f(x) dx$
- *Variance and Standard Deviation (measure of variability/uncertainty):*

$$\mathbb{V}_X = \int_{x \in \mathcal{D}_X} f(x)(x - \mathbb{E}_X)^2 dx, \qquad \mathbb{SD}_X = \sqrt{\mathbb{V}_X}$$

# Summarizing Statistics (Sample Approximations)

Need theoretical pdf of $X$ to compute expected value, variance, and SD (theoretical statistics).

However, if we have data $x_\omega$, $\omega \in \mathcal{S}$, we can approximate theoretical statistics by using empirical estimates:

- *Empirical Mean:*

$$\hat{\mathbb{E}}_X = \frac{1}{S} \sum_{\omega \in \mathcal{S}} x_\omega$$

- *Empirical Variance and Standard Deviation:*

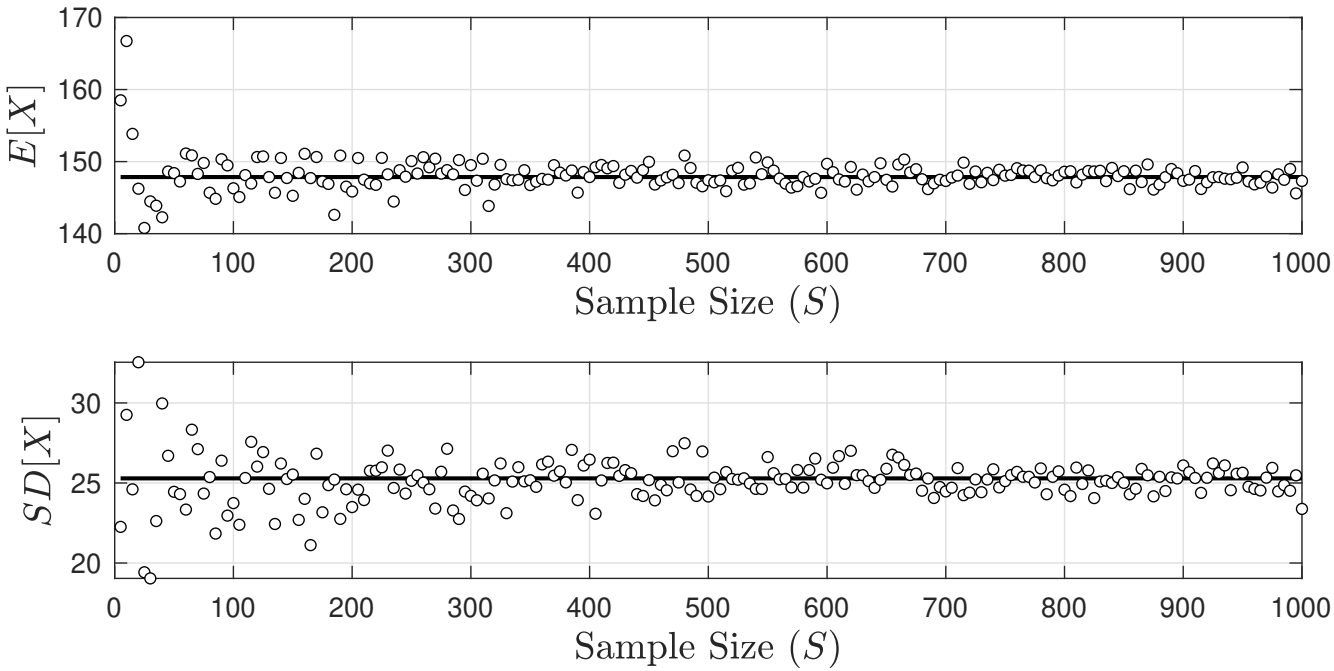
$$\hat{\mathbb{V}}_X := \frac{1}{S} \sum_{\omega \in \mathcal{S}} (x_\omega - \hat{\mathbb{E}}_X)^2, \qquad \hat{\mathbb{SD}}_X = \sqrt{\hat{\mathbb{V}}_X}$$

Intuition tells us that approximations improve as we accumulate data (as $S$ becomes large). We will see later that this is indeed the case.

Statistics can be related to model parameters (e.g., for a Gaussian: $\mathbb{E}_X = \mu$ and $\mathbb{V}_X = \sigma^2$). As such, empirical estimates of statistics can be used to obtain parameters.

# Example: Gibbs Reactor ch1_gibbs_example [matlab] [live] [python]

Use reactor pressure data to compute empirical estimates for the mean $\hat{\mathbb{E}}_X$ and standard deviation $\hat{\mathbb{SD}}_X$ (we explore effect of using increasing amounts of data $S$).




Empirical mean and SD converge to the theoretical values $\mathbb{E}_X = 148$ and $\mathbb{SD}_X = 25$

Estimates are not accurate for small sample sizes.

# Summarizing Statistics (Quantiles)

An important family of summarizing statistics are the quantiles (a.k.a. percentiles).

- The quantile is the inverse function of the cdf and, as such, it might be easier to explain it from this perspective.

- Consider the following equation for some $\alpha \in [0, 1]$:

$$F_X(x) = \mathbb{P}(X \leq x) = \alpha$$

- A value $x$ that satisfies equation is the $\alpha$-quantile of RV $X$ and is denoted as $\mathbb{Q}_X(\alpha)$.

- This means that we can express the quantile as:

$$\mathbb{Q}_X(\alpha) = F_X^{-1}(\alpha)$$

# Summarizing Statistics (Quantiles)

Some important observations about quantiles:

- Since cdf can have a "staircase" form, there might be multiple values of $x$ satisfying $F_X(x) = \alpha$. Consequently, $\alpha$-quantile might not be unique.
- Typically, the definition of the quantile is refined by looking for the smallest or center values of $x$ satisfying $F_X(x) \geq \alpha$.
- Quantiles are related to other summarizing statistics of interest. For instance:
  - $\mathbb{Q}_X(0.5)$ is the *center value* of $X$ (a.k.a. the median and denoted as $\mathbb{M}_X$)
  - $\mathbb{Q}_X(1) = \max_{x \in \mathcal{D}_X} x$ is the maximum value of $X$
  - $\mathbb{Q}_X(0) = \min_{x \in \mathcal{D}_X} x$ is the minimum value of $X$
- We can use empirical cdf $\hat{F}_X(x)$ to estimate empirical quantiles $\hat{\mathbb{Q}}_X(\alpha)$.

# Summarizing Statistics (Mode and Moments)

- The mode of an RV is the outcome of maximum probability:

$$\mathbb{MO}_X \in \underset{x \in \mathcal{D}_X}{\operatorname{argmax}} f_X(x).$$

  Some RVs are unimodal (one peak) and some are multimodal (multiple peaks).

- *Central moments* are an important family of summarizing statistics
  - The central moments of $X$ are given by:

$$\mathbb{CMO}_k[X] = \mathbb{E}[(X - \mathbb{E}[X])^k], \qquad k = 1, 2, 3, 4, ...,$$

    Note that the second ($k = 2$) central moment is the variance.

  - The standardized moments of $X$ are given by:

$$\mathbb{SMO}_k[X] = \frac{\mathbb{E}[(X - \mathbb{E}[X])^k]}{\mathbb{SD}[X]^k}, \qquad k = 1, 2, 3, 4, ...,$$

    The third ($k = 3$) standardized moment is known as *skewness* and the fourth ($k = 4$) is known as *kurtosis*. Skewness is a measure of the asymmetry of the pdf, while kurtosis is a measure of the nature of the tails of the pdf.

# Summarizing Statistics (Mode and Moments)

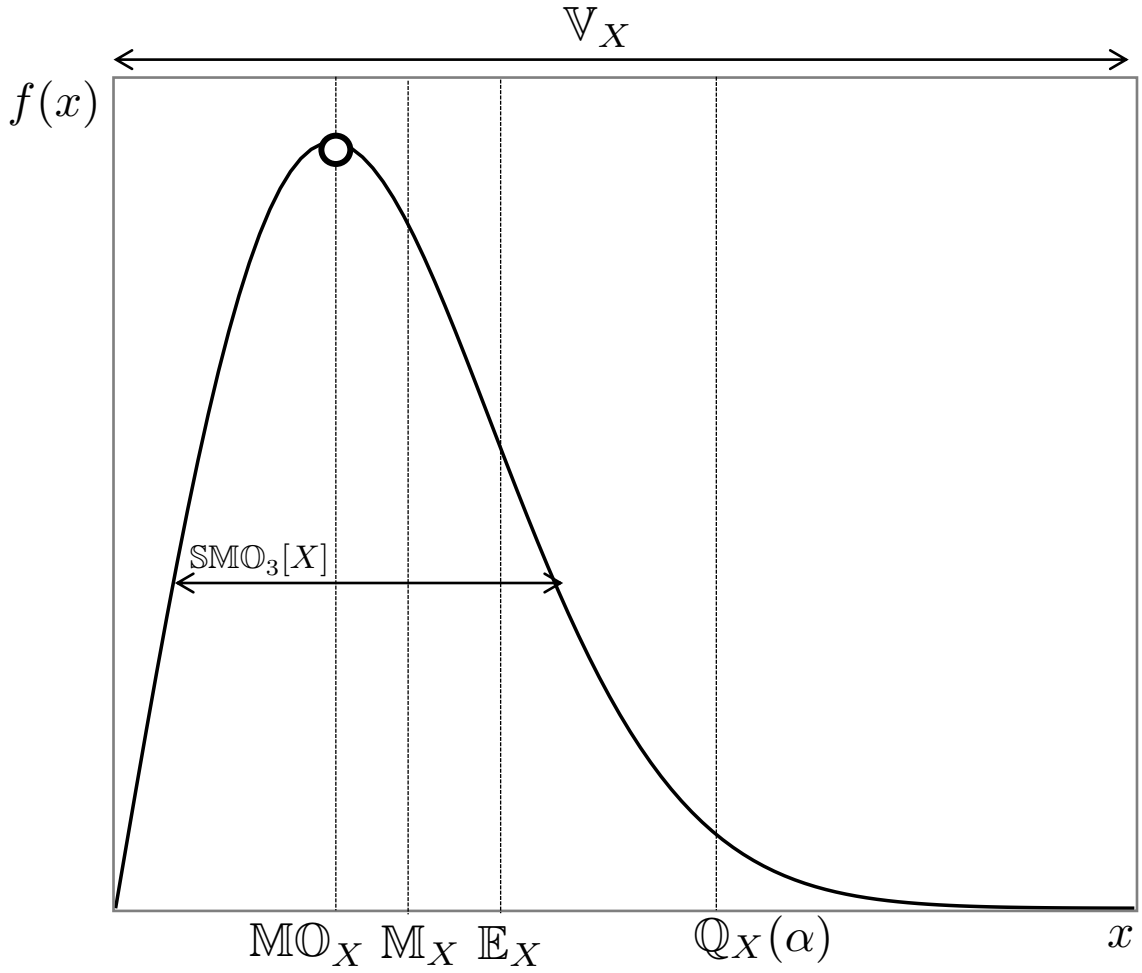



Figure: Features of the probability density function that different summarizing statistics capture.

# From Knowledge to Decisions

We now have a characterization of a given random phenomenon (our RV model $X$) affecting a system of interest and we would like to exploit this to make decisions.

In this context, it is important to make a couple of observations:

- Uncertainty *propagates* through systems in complex ways.
- Uncertainty can be *mitigated* via design or control decisions.

# Uncertainty Propagation and Mitigation

- Consider propagation of $X$ through system $\varphi(X, u)$:

$$Y = \varphi(X, u)$$

  where $u$ is a mitigating action (decision) and $Y$ is the system output.

- We make the following observations:

- Output $Y$ is an RV if the input $X$ is an RV.
- Nature of $Y$ (its cdf, pdf, and domain) depends on system $\varphi$. Some systems magnify uncertainty and variability while others might damp it.
- Nature of $Y$ depends on action $u$. Can use action to mitigate/manipulate uncertainty of $Y$.

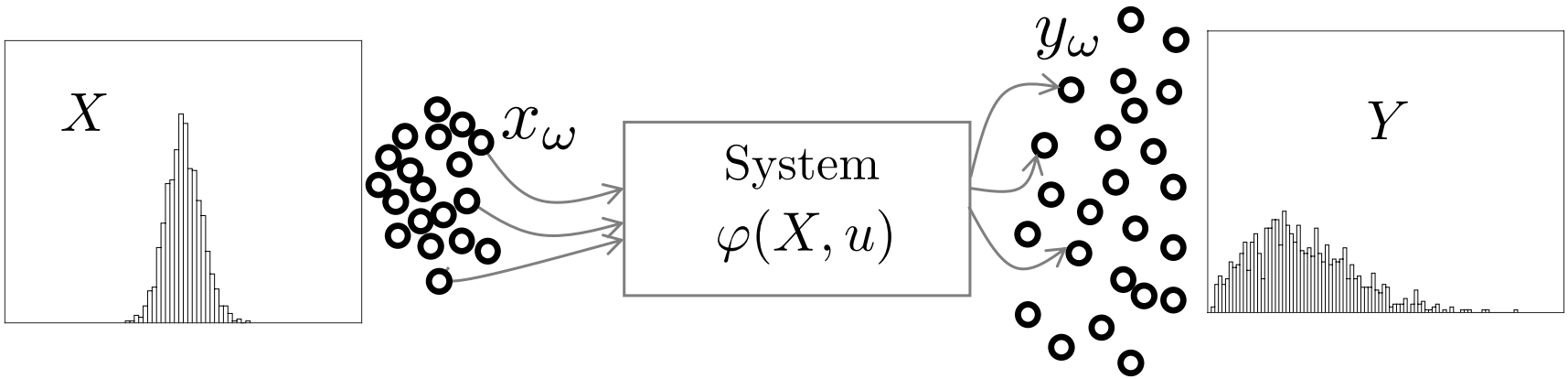



Figure: Illustration of propagation of a random variable $X$ through a system. The propagation results in a random output $Y$ that has different characteristics (e.g., higher variability/uncertainty).

# Uncertainty Propagation and Mitigation

Having data $x_\omega$, $\omega \in \mathcal{S}$ and a system model $\varphi$, we can characterize cdf, pdf, domain, and summarizing statistics of $Y$ using the following simulation procedure:

1. For a given decision $u$, perform simulations of the form:

$$y_\omega = \varphi(x_\omega, u),\ \omega \in \mathcal{S}$$

2. Use $y_\omega$ to compute sample approximations of quantities of interest for $Y$ such as:
   - Sample mean:

$$\hat{\mathbb{E}}_Y = \frac{1}{S}\sum_{\omega\in\mathcal{S}} y_\omega = \frac{1}{S}\sum_{\omega\in\mathcal{S}} \varphi(x_\omega, u)$$

   - Sample variance:

$$\hat{\mathbb{V}}_Y = \frac{1}{S}\sum_{\omega\in\mathcal{S}} (y_\omega - \hat{\mathbb{E}}_Y)^2$$

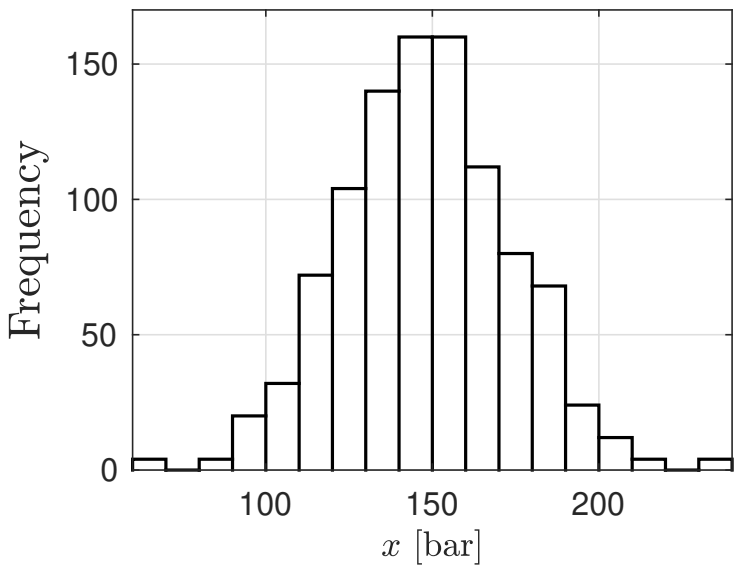
   - Empirical cdf:

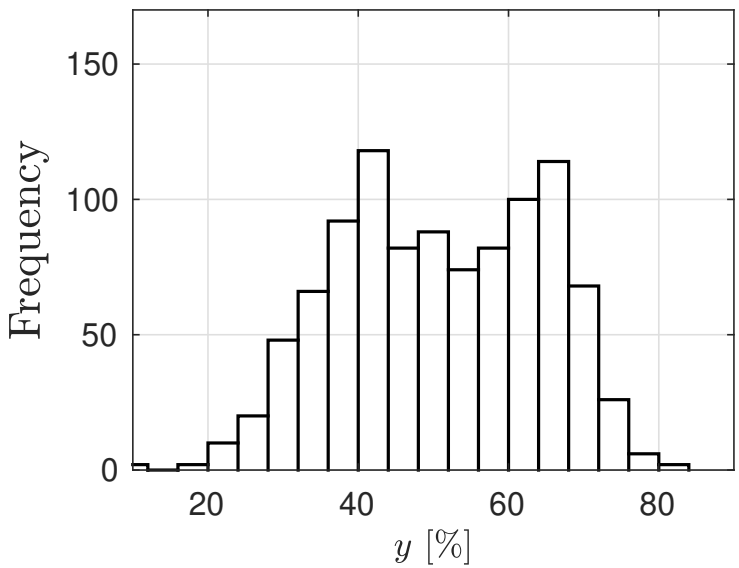
$$\hat{F}_Y(y) = \frac{1}{S}\sum_{\omega\in\mathcal{S}} \mathbf{1}[y_\omega \le y]$$

The above procedure is known as *Monte Carlo (MC) simulation* and is widely used to estimate diverse quantities of interest for RVs.

# Example: Gibbs Reactor `ch1_gibbs_example` [matlab] [live] [python]

- Empirical pdf and cdf for pressure (input $X$) and conversion (output $Y$).
- Note change in behavior of $Y$; pdf of $X$ is unimodal, while pdf of $Y$ is bimodal.




- Complex behavior of the conversion pdf is the result of strong nonlinear behavior

# Decision-Making under Uncertainty

We would like now to find a decision $u$ that manipulates $Y(u) = \varphi(X, u)$ in some desirable way (e.g., minimizes uncertainty/variance).

Consider the questions:

- If we have a couple of competing decisions $u$ and $u'$ giving rise to random outputs $Y(u)$ and $Y(u')$, how can we tell which one is better?
- How can we find the best possible decision $u$? What do we mean by the "best"?

# Decision-Making under Uncertainty

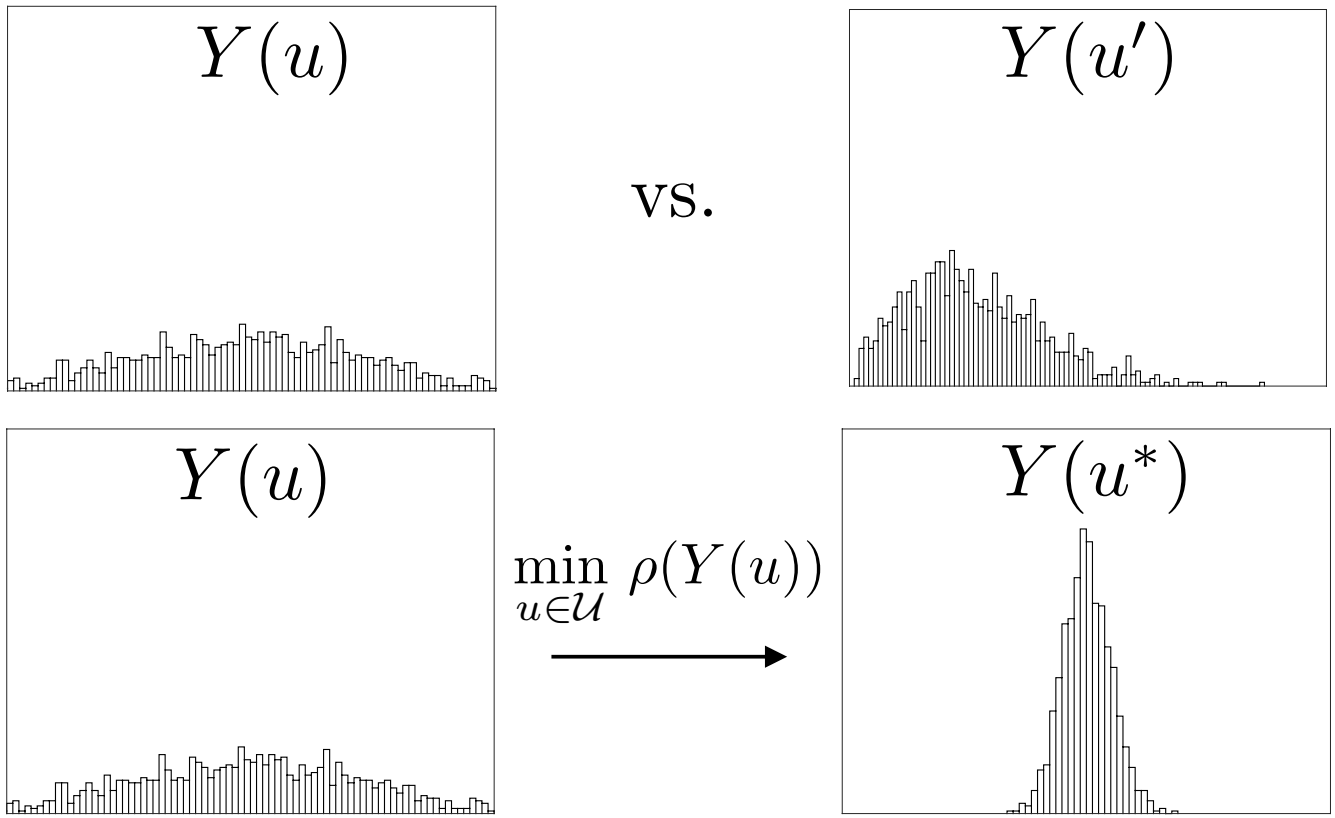



Figure: Paradigms for decision-making under uncertainty. Comparison between decisions $u, u'$ and associated outputs $Y(u), Y(u')$ (top). Find best decision $u^*$ that shapes $Y(u^*)$ in a desirable way (bottom).

# Decision-Making under Uncertainty

- If we assume *deterministic setting* (no uncertainty), then $Y(u)$ and $Y(u')$ will each take a single value $y(u)$ and $y(u')$ and one would select, *unambiguously*, the one with smaller (or larger) value. For instance, one would select $u$ over $u'$ if $y(u) \leq y(u')$.

- In a *setting under uncertainty* this is no longer possible because $Y(u)$ and $Y(u')$ have multiple possible outcomes ($Y(u)$ and $Y(u')$ are RV models)

- Concept of "better" under uncertainty is ambiguous and the mathematical statement $Y(u) \leq Y(u')$ does not even make sense.

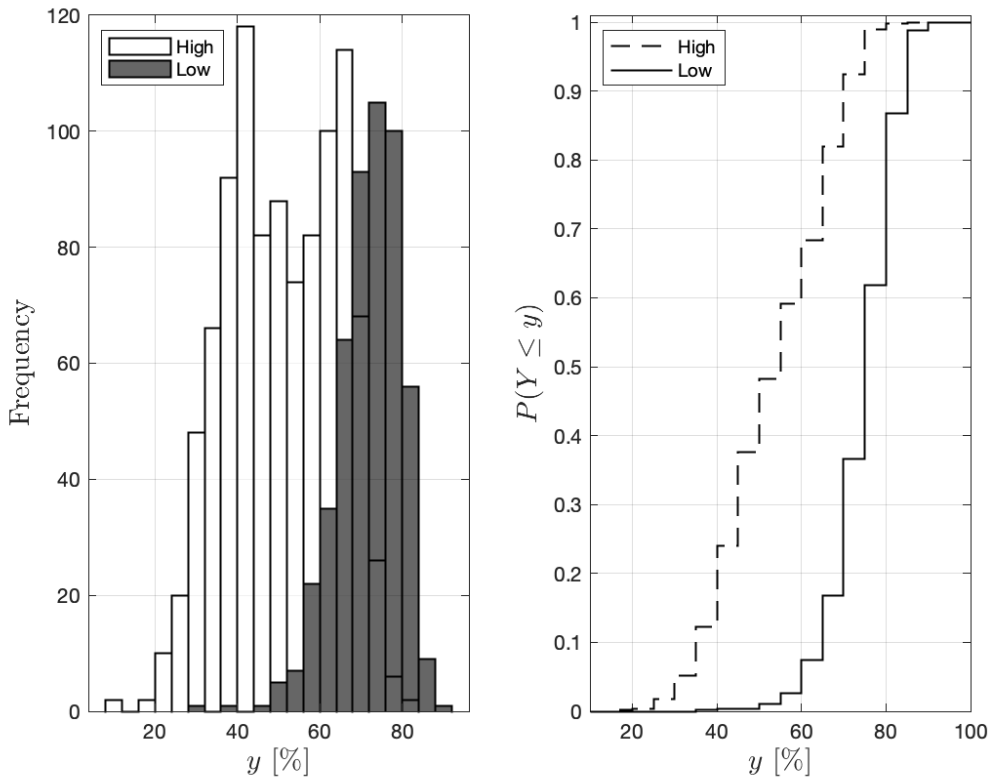
- Does $Y(u) \leq Y(u')$ mean that all outcomes of $Y(u)$ are lower than those of $Y(u')$? Does it mean that a subset of outcomes are lower?

- When making a decision under uncertainty, we need to capture information embedded in $Y(u)$ and $Y(u')$.

- This requires comparing RVs consistently; e.g., by using their cdfs or by using *risk measures* (summarizing statistics).

# Example: Gibbs Reactor `ch1_gibbs_example` [matlab] [live] [python]

- Can counteract variability in pressure $X$ by modifying the operating temperature $(u)$.
- We compare empirical pdf and cdf for conversion at low $(Y(u))$ and high $(Y(u'))$ temperature
- Should we operate at low or high temp?
- By comparing cdfs, we can see that operating at low temp is *consistently* more likely to achieve higher yields.



# Decision-Making under Uncertainty

- We might not only be interested in comparing decisions, but we might want to find the *best* possible decision.
- To decide what is "best", we select a measure of the output $Y(u)$ (a summarizing statistic) that we denote as $\rho(Y(u))$.
- We find best decision by solving optimization problem:

$$\min_{u \in \mathcal{U}} \ \rho(Y(u)).$$

- $\mathcal{U}$ is set of possible decisions that we can choose from. In Gibbs reactor example, $\mathcal{U} = \{583, 613\}$.
- If we select $\rho(Y(u)) = \mathbb{SD}[Y(u)]/\mathbb{E}[Y(u)]$, optimization problem finds decision $u$ such that $Y(u)$ minimizes the *coefficient of variation* (CV).
- In Gibbs reactor, operating at a low temp yields a CV of $0.07$, while operating at high temp yields $0.10$ (optimal decision is $u^* = 583$ K).
- We will see later that $\rho(Y(u))$ is a risk measure that aims to model attitudes towards risk by decision-makers.